Clearbrook and SGS Playwright Convergence

Executive Summary

Status: Draft for joint discussion
Purpose: Establish agreement on the target direction, primary deliverables, and workflow completion criteria

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1. Proposed direction

Clearbrook and SGS have made meaningful progress toward a combined Playwright automation solution. The proposed direction is to complete that convergence through one shared platform and delivery model:

- **`PnC-TestAutomation` is the target platform.**
- The Clearbrook and SGS repositories will not remain independent long-term solutions.
- SGS workflows will be adapted to the target platform rather than supported through a separate framework.
- Selected SGS framework capabilities will be incorporated where they strengthen the combined solution.
- There will be **one CI/CD path** for packaging, execution, and result handling.
- There will be **one reporting and evidence path**, including Azure Test Plans integration.
- Clearbrook will govern the shared framework and enterprise delivery standards.
- SGS will remain responsible for adapting and completing the contracted Duck Creek workflows.

The objective is not to discard the SGS work. It is to preserve its workflow coverage and useful capabilities while moving delivery into a single, supportable automation platform.

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2. SGS capabilities evaluated for adoption

The following SGS capabilities have been evaluated for selective adoption, alignment, or reimplementation within the combined framework:

- Centralized control actions, waits, diagnostics, and verification
- Deferred verification with evidence produced before final failure reporting
- Duck Creek frame-aware locator resolution
- Layered scenario-data resolution and validation
- Scenario-level evidence management and deterministic cleanup
- Dropdown and option-selection behavior that rejects ambiguous matches

Several of these concepts have already influenced or been incorporated into the combined framework. Final use will be confirmed through implementation and workflow qualification rather than by transferring the SGS framework wholesale.

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3. SGS deliverables

SGS will lead the product-specific conversion and adaptation work needed to bring the contracted Duck Creek workflows into the target platform:

1. **Tosca workflow conversion and traceability** - Map each converted workflow to its approved source test case. - Preserve required business paths, branches, and expected outcomes.
2. **Workflow adaptation** - Adapt converted workflows to the approved lifecycle, controls, data, evidence, and reporting services. - Remove dependencies on a separate SGS execution framework.

3. **Locator and frame corrections** - Correct product-specific locators, control mappings, and Duck Creek frame handling. - Resolve ambiguous or screen-specific behavior for the supported products and states.
4. **Data-profile cleanup** - Confirm required workflow inputs and correct missing, copied, inconsistent, or environment-specific values. - Maintain traceability between the test case, data profile, and execution.
5. **Business-outcome validation** - Implement the required checks for selected values, quote identity, premium, forms, referrals, corrections, transmission, and other applicable outcomes.
6. **Test Plans integration support** - Provide and maintain authoritative mappings between converted workflows and Azure Test Plans cases. - Support result and evidence reconciliation.
7. **Packaging and pipeline updates** - Supply all required SGS workflow assets. - Align migration-specific packaging, execution, and reporting components with Clearbrook standards.
8. **Platform coverage** - Execute the agreed browser and device coverage for applicable workflows using Clearbrook-provided access.
9. **Product documentation and knowledge transfer** - Document workflow behavior, data requirements, mappings, known constraints, and troubleshooting guidance. - Transfer sufficient product knowledge for Clearbrook to operate and support the converted workflows.

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4. Clearbrook deliverables

Clearbrook will lead the shared platform, governance, and enterprise delivery capabilities:

1. **Framework governance** - Define and maintain the target architecture, shared interfaces, contribution standards, and technical direction.
2. **Shared framework services** - Maintain the common lifecycle, controls, data resolution, business-outcome, evidence, reporting, and integration services.
3. **Internal framework defect remediation** - Correct defects in the Clearbrook-owned framework that prevent successful workflow adaptation or execution.
4. **CI/CD ownership** - Operate the approved pipelines, agents, identities, schedules, execution controls, and result-publication path.
5. **DevOps standards** - Define the package, pipeline, environment, promotion, failure-handling, and operational requirements.
6. **Security architecture** - Define protected-configuration, identity, transport, access, evidence-handling, and data-protection requirements.
7. **Expected business outcomes** - Confirm required product and state coverage, scenario intent, expected results, and acceptable tolerances.
8. **Environment and access dependencies** - Provide the agreed application environments, test identities, execution agents, device or browser access, and Azure Test Plans permissions.
9. **Review and operational readiness** - Review technical alignment and business outcomes. - Confirm that the combined solution can be operated and supported through the agreed delivery path.

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5. Proposed workflow acceptance criteria

A converted workflow is ready for acceptance when it:

1. **Executes successfully and repeatably** - Completes the intended business path without unresolved errors or required manual intervention.
2. **Uses the approved framework** - Uses the agreed architecture, interfaces, and shared framework services rather than introducing a separate runtime or duplicate capability.

3. **Produces the expected business outcomes** - Applies the approved inputs and verifies the required product and downstream results.
4. **Runs through the agreed delivery path** - Executes from the approved package through the shared pipeline.
5. **Produces the required result and evidence** - Reports one accurate outcome with the evidence needed to understand successful and failed execution. - Publishes to Azure Test Plans when required.
6. **Meets the agreed runtime target** - The average Playwright execution duration is equal to or better than the approved average Tosca duration for the equivalent workflow.
7. **Is supportable** - Required documentation, mappings, dependencies, and known limitations are current and available to both teams.

Runtime comparison method

To ensure a fair and repeatable comparison:

- The Playwright and Tosca executions must represent the same business workflow and expected outcomes.
- The comparison must use the same or equivalent environment and data conditions.
- The approved Tosca baseline must be recorded before evaluating the converted workflow.
- The result must be based on at least five comparable successful executions.
- Warm-up runs, infrastructure failures, and runs containing retries are excluded.
- Active workflow execution must be measured consistently; pipeline queue, agent provisioning, and package download time are reported separately.
- Both average and median duration will be recorded to make environmental outliers visible.
- Runtime improvement cannot be achieved by removing required validations or evidence.
- Any environmental tolerance will be agreed before the comparison is performed.

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6. Agreement requested

Clearbrook and SGS should use the next alignment session to confirm:

- `PnC-TestAutomation` as the target platform;
- convergence into one framework rather than independent repositories;
- the SGS capabilities to incorporate into the shared platform;
- the SGS and Clearbrook deliverable boundaries;
- one CI/CD, reporting, evidence, and Test Plans path;
- the workflow acceptance criteria and Tosca runtime comparison method; and
- the first prioritized workflow increment, owners, dependencies, and target dates.

Once this direction is agreed, the detailed working matrix and delivery backlog can be used to manage implementation.